Stable matching with hidden options

Auditing access in matching markets

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The question

Suppose a broker shows a client only some of their feasible options. A better match may never reach them.

We ask which firms a platform should audit to make its matching stable, including against the options that were initially hidden.

The difficulty is that opening an option can change the matching algorithm's choice.

We model access to opportunities. Commissions and brokers' reasons for hiding options are outside the model.

The market

Buyers B and sellers S, with one property per seller. Each client can have at most one partner.

$$E \subseteq B \times S \qquad \text{is the set of mutually acceptable, feasible pairs.}$$

Each client strictly ranks its feasible partners. Every acceptable partner is better than being unmatched, written $\perp$.

Each client i has one fixed representative $f(i)$. A firm can represent many clients. A *side-exclusive* firm represents only buyers or only sellers.

Prices, rankings, and representation stay fixed. Brokers control access. For rentals, interpret the two sides as renters and listing slots.

Stability in the full market

For a matching M, let $M(i)$ be client i's partner, or $\perp$ if unmatched.

$$\mathrm{BP}_E(M) = \{bs \in E \setminus M : s \succ_b M(b) \text{ and } b \succ_s M(s)\}.$$

A pair blocks when both clients would rather match with each other.

We call M *full-market stable* when $\mathrm{BP}_E(M) = \varnothing$.

Deferred acceptance is stable on the graph it receives. A pair left out of that graph can still block in E.

This treatment of unavailable pairs follows restricted-edge matching (Cseh and Manlove, 2016).

Audits and deferred acceptance

Initially, only $E_0 \subseteq E$ is available. Let $\mathcal{F}$ be the firms. Auditing a set $A \subseteq \mathcal{F}$ opens

$$E(A) = E_0 \cup \{bs \in E : f(b) \in A \text{ or } f(s) \in A\}.$$

Either representative is enough. An audit enables all feasible pairs involving its clients, without forcing a transaction.

$$c(A) = \sum_{f \in A} c_f, \quad c_f \in \mathbb{Z}_{>0}, \qquad D(A) = \mathrm{DA}_B(E(A)).$$

Deferred acceptance (DA): buyers propose down their available lists. Each seller keeps its favorite proposal so far and rejects the others. Rejected buyers continue until no further proposal is possible.

Assumption: the planner knows E and the rankings. Payment enables access. The platform always uses buyer-proposing DA.

Two audit costs

Response: DA must select a full-market stable matching.

$$C_{\mathrm{DA}} = \min_{A \subseteq \mathcal{F}} \{c(A) : \mathrm{BP}_E(D(A)) = \varnothing\}.$$

Support: some full-market stable matching must be available.

$$C_{\mathrm{support}} = \min_{A \subseteq \mathcal{F}} \{c(A) : \exists M \subseteq E(A),\ \mathrm{BP}_E(M) = \varnothing\}.$$

Every successful DA response supplies such a matching, so

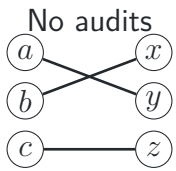
$$C_{\mathrm{support}} \leq C_{\mathrm{DA}}.$$

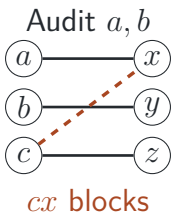
Auditing every firm is feasible. Running DA is polynomial-time. Choosing the audits is the optimization problem.

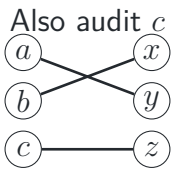
A small example

Each client has its own firm. Every pair is feasible. Initially $E_0 = \{ay, bx, cz\}$.

$$a : x \succ y \succ z \quad b : y \succ x \succ z \quad c : x \succ z \succ y$$
$$x : b \succ c \succ a \quad y : a \succ b \succ c \quad z : c \succ a \succ b$$







Solid lines are assignments. The dashed pair is hidden. All three stability checks use the same full graph and the same preferences.

The rejection chain

After auditing a, b, both get their first choices. But c still only sees z. Now c prefers x, and x prefers c to its new partner a.

After auditing c, restart DA. Let a, b propose first, then c:

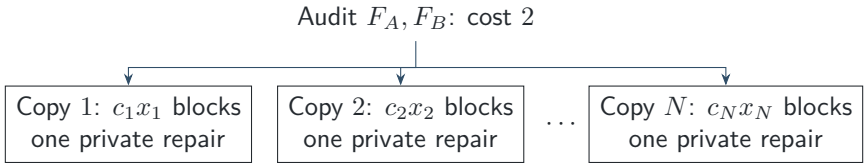
Proposal	Listing keeps	Displaced client
c to x	c	a
a to y	a	b
b to x	b	c
c to z	c	none

We return to the original stable matching. Adding audits can turn success into failure. Success is not monotone.

Two audits versus $N + 2$

Theorem. For every $N \geq 1$, unit costs can give $C_{\text{support}} = 2$ and $C_{\text{DA}} = N + 2$.

Take N copies of the example. All a_i share firm F_A, all b_i share F_B, and other firms are private. Each copy starts at its stable matching.



Each shared firm also represents $N + 3$ buyers in separate hidden one-edge markets. Their sellers have private firms. These make substitutes for F_A, F_B expensive.

Unit costs and side-exclusive firms. The shared portfolios grow with N.

Why the gap is exact

1. **Support costs two.** F_A, F_B open every isolated pair. The original stable matchings remain available. One audit cannot open both isolated-pair families.

2. **A cheapest repair uses both shared firms.** Omitting F_A requires $N + 3$ private seller audits, plus one for F_B's family: at least $N + 4$. The same holds for omitting F_B.

3. **Each copy costs one more.** With F_A, F_B audited, we must audit $f(c_i)$ or $f(x_i)$. Either exposes $c_i x_i$ and repairs that copy.

Those N private choices are disjoint. Auditing F_A, F_B and all $f(c_i)$ achieves $N + 2$, so the bound is tight.

Appendix 1 gives a gap of 1 versus $N + 1$ with one client per firm. This is a cost separation, not an approximation-hardness theorem.

A two-by-two switch

For an edge $e = \{u, v\}$ in a graph, use buyers p, q represented by firms F_u, F_v. Sellers x, y have private firms. Initially expose px, qx.

$$p : y \succ x, \quad q : x \succ y, \quad x : p \succ q, \quad y : q \succ p.$$

Relevant audit	DA outcome	Full-market stability
None	px, with q, y unmatched	qy blocks
F_u	py, qx	stable
F_v alone	px, qy	stable
y's firm	py, qx	stable

Success is exactly $F_u \in A$ or $F_v \in A$ or $f(y) \in A$.

A reduction from Vertex Cover

Vertex Cover selects vertices touching every edge. Use degree ≤ 3.

Graph	Matching instance
Vertex v	Buying firm F_v
Edge $\{u, v\}$	One switch using firms F_u, F_v
Degree of v	Number of clients represented by F_v

A cover audits an endpoint of every switch. To go back, replace private y-firm audits by endpoint firms. The OR condition preserves success without increasing cost.

The optima coincide: NP-complete budget decision, APX-hard optimization. This holds with at most three clients per firm and two-item lists.

Unit costs and side-exclusive firms. Verification is polynomial. Degree-three Vertex Cover: Garey et al. (1976), Alimonti and Kann (2000).

The algorithm

1. Find the cheapest plan that supports a full-market stable matching, subject to constraints learned so far. This is the *master problem*.
2. Run DA on that plan. Check for blocking pairs in the full market.
3. If a pair blocks, add a constraint excluding that failure. Solve again.
4. Stop when DA passes, or the lower bound reaches the cost of a previously verified plan.

Logic-based Benders decomposition (Hooker and Ottosson, 2003), using the stable-matching polytope (Rothblum, 1992). Equations: Appendix 2.

The part of each list that matters

Suppose a buyer ranks $x \succ y \succ z$ and DA assigns it to y. Its *prefix* contains x and y, even if x was hidden.

For a visible graph $H \subseteq E$, let $M = \mathrm{DA}_B(H)$. The full prefixes are

$$\Pi(M) = \{bs \in E : M(b) = \bot \text{ or } s \succeq_b M(b)\}.$$

An unmatched buyer contributes its whole list.

DA never proposes below its eventual partner. Keep every prefix pair visible or hidden exactly as before, and the same DA run replays.

To fix a failed outcome, at least one of these visibility choices must change.

Proof: Appendix 3. Related DA-prefix reasoning appears in Aguiar and Hong (2026, preprint).

Constraints from a failed run

Constraint	What it rules out
Whole assignment	Only this complete vector of firm audits.
Component assignment	Every plan keeping the failed component's firm decisions.
Component prefix	Every plan preserving the relevant local exposure statuses.

For the three-by-three failure, let v_e indicate whether pair e is visible. A valid constraint is

$$(1 - v_{ax}) + (1 - v_{by}) + v_{cx} \geq 1.$$

At $(v_{ax}, v_{by}, v_{cx}) = (1, 1, 0)$, the left side is zero: this pattern fails.

Changing a term is necessary, not sufficient. We choose the whole audit plan before implementation, so a revised plan can omit an optional audit.

What the solver certifies

The master gives a lower bound L. A plan that passes the DA check gives an upper bound U:

$$L \leq C_{\mathrm{DA}} \leq U.$$

Every failure excludes a new audit pattern. With exact master solves, the algorithm eventually certifies an optimum. A timeout leaves a verified plan and a gap $U - L$.

On the shared-firm construction, prefix cuts need at most two master solves. Whole-assignment cuts need at least $2^N - 1$ failed DA checks.

This round bound compares these cut methods. It is not a polynomial-time guarantee. Proofs: Appendices 4–5. Numerical certification uses solver tolerances.

Synthetic markets

The core has 24 clients per side and ten side-exclusive firms, with unit audit costs. Pairs are feasible within each local market and absent between markets.

Factor	Values
Local markets	1 or 4
Rankings	independent or correlated
Mean initial exposure	25% or 50%
Shared seed blocks	20

$2 \times 2 \times 2 \times 20 = 160$ markets. Every method sees the same instances. Partnerships increase exposure odds. A random half of buyers receive lower exposure odds, independently of preferences.

All results are synthetic. Lower access is randomly assigned, not a demographic category. Bootstrap intervals resample whole seed blocks.

Methods

Method	How it chooses audits
Three exact variants	Same support model and starting heuristic, different failure cuts.
Exhaustive search	Check all 2^{10} firm-audit vectors.
Reverse-delete	Start with all firms. Keep a deletion only if DA still passes.
Blocker greedy	Add the audit with greatest immediate blocker reduction per cost.
Prefix cover	Expose all prefixes of the full-market DA outcome.
Full disclosure	Audit every firm.

Every returned plan passes the DA check. The heuristics have no approximation guarantee here. The prefix-cover counterexample is in Appendix 6.

Results on 160 markets

Method	Mean cost	At min.	Median (ms)
Exhaustive search	4.74	160	20.8
Assignment cuts	4.74	160	55.3
Component cuts	4.74	160	54.7
Prefix cuts	4.74	160	50.8
Reverse-delete	4.87	140	1.4
Blocker greedy	5.56	62	3.9
Prefix cover	4.98	126	7.4
Full disclosure	10.00	0	—

All exact variants certified every optimum. Reverse-delete usually matched it.

"At min." counts the markets attaining the known optimum (out of 160). Paired prefix/assignment runtime ratio: 1.007, 95% interval [0.995, 1.021]. These data show no general speed advantage for prefix cuts.

The support lower bound

What happens if we remove the support model? Keep prefix cuts and the same starting heuristic. Use 40 core markets and a five-second cap.

Model	Certified optima	Median seconds
With support	40/40	0.046
Without support	18/40	5.003

With support, prefix cuts certified 151 of the full 160 markets after one master solve.

The support optimum was strictly below the DA optimum in eight markets, by one audit each. The other 152 markets had equal optimum costs.

The 40 markets are a subset of the core sample. The capped median includes timeouts.

Larger markets

64 clients per side, 24 firms, and a fifteen-second cap. There are 16 markets in each row below.

Regime	Certified	Median (s)	Max. $U - L$
One connected market	9/16	9.549	2
Eight local markets	16/16	0.061	0

These are prefix-cut results. All three exact variants certified the same cases. The gap $U - L$ is in audit-cost units.

Seven connected cases remained unresolved. A timeout does not tell us whether the returned plan is suboptimal.

Every returned plan passed the stability check. Splitting markets also shortens lists, so the comparison does not isolate the effect of decomposition.

Calls needed to certify the optimum

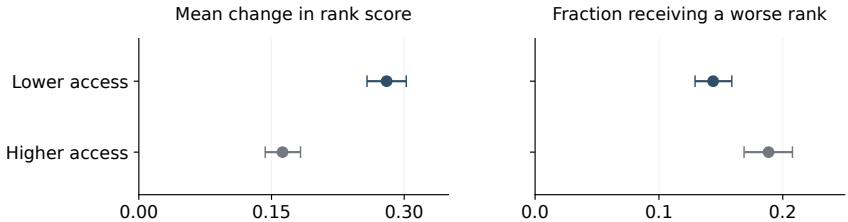
Number of three-by-three copies: N. Time limit: 10 seconds.

N	Assignment cuts	Component cuts	Prefix cuts
1	10	3	2
2	131	4	2
3	*Timeout*	4	2
4	*Timeout*	5	2
5	*Timeout*	5	2
6	*Timeout*	4	2
8	*Timeout*	7	2
10	*Timeout*	6	2

Timeout: no certificate within 10 seconds. A feasible plan was still returned. Calls count master solves, excluding initialization.

Who gains, and who loses?

Rank score: 0 if unmatched, otherwise $1 - (\text{rank} - 1)/(\text{list length})$.



Among lower-access buyers, the selected plans improve mean rank score, but 14.4% receive a worse-ranked partner than in the initial DA outcome.

160 synthetic markets, prefix solver's selected optimal plans. Intervals are 95% seed-block bootstrap intervals. Rank score is an ordinal summary, not welfare. The initial allocation is provisional.

Different optimal plans

We enumerated all 614 cost-optimal audit plans in the core. 148 markets had multiple optimal plans. In 37, those plans produced different allocations.

Every optimal plan worsens at least one...	Markets
buyer's rank	155/160
seller's rank	159/160
lower-access buyer's rank	135/160

A different cheapest plan can change who loses. It often cannot avoid individual losses altogether.

Losses are relative to the initial provisional DA assignment. These counts do not say that every stable intervention has the same losses.

Housing and AI recommendations

A housing recommender could determine which feasible options reach a client. In our model, those options form E_0.

The audit problem would apply to a service that can open additional options and uses DA to assign clients. We have not established that a housing platform operates this way.

Our experiments measure the effects of synthetic access restrictions. They do not evaluate an AI recommender or a real housing intervention.

Liu et al. (2024) audit LLM housing recommendations for racial steering. A proposed listing study and access-cost objective appear in Appendices 14–15.

Related work

Work	Connection to this project
Cseh–Manlove (2016)	Unavailable pairs can still block.
Rastegari et al. (2016)	Costly interviews and full-information benchmarks.
Aguiar–Hong (2026)	Limited attention, discovery, and DA-prefix reasoning.
Boehmer et al. (2021)	Interventions in stable-matching markets.
Trabelsi et al. (2025)	Costly pair additions under a fixed maximum-matching rule, with participation guarantees.

We study grouped access decisions with a fixed DA response. Success requires stability against the unchanged full market.

We use established DA, stable-polytope, and Benders methods. Aguiar–Hong is a preprint. Full references are in the appendix.

Conclusions

Making a stable matching available can cost much less than making DA select one.

Choosing the audits is NP-hard. Our exact method learns constraints from failed DA runs and reports a verified plan and a cost bound.

On the synthetic markets, simple heuristics are competitive. Even a cheapest successful plan can leave some clients worse off.

The housing application needs a service whose access rules and matching process fit the model.

Stability is a guarantee about the input rankings. It does not guarantee fair rankings, truthful preferences, or improvements for every client.

1. A simpler gap with one client per firm

Start each of N independent copies with

$$H = \{ax, ay, az, bx, by, bz, cz\}.$$

$\{ay, bx, cz\}$ is already supported and full-market stable. DA instead returns $\{ax, by, cz\}$, blocked by cx.

Without auditing c or x, the only possible addition is cy. It changes nothing, since c keeps z. Auditing either c or x repairs the copy.

Add one hidden isolated pair. With unit costs and private firms,

$$C_{\text{support}} = 1, \qquad C_{\text{DA}} = N + 1.$$

The shared-firm construction also shows how audits needed in one part of the market can destabilize another part.

2. The support master in equations

$x_f \in \{0,1\}$ chooses audits. Exposure $v_{bs} \in [0,1]$ obeys

$$v_e = 1 \ (e \in E_0), \qquad v_{bs} \geq x_{f(b)}, x_{f(s)}, \quad v_{bs} \leq x_{f(b)} + x_{f(s)} \ (bs \notin E_0).$$

Let $\delta(i)$ denote incident pairs. Minimize $\sum_f c_f x_f$ with $z_e \geq 0$:

$$\sum_{e \in \delta(i)} z_e \leq 1 \qquad \text{for every client } i,$$

$$z_{bs} + \sum_{s' \succ_b s} z_{bs'} + \sum_{b' \succ_s b} z_{b's} \geq 1 \qquad bs \in E,$$

$$z_e \leq v_e \qquad e \in E.$$

For fixed integral x, v is integral. Setting hidden z_e to zero gives a face of the integral stable-matching polytope, so continuous z is enough.

Rothblum (1992). This argument is for a fixed integral audit vector. It does not assert integrality when the audit variables are fractional.

3. Why the prefix constraint is valid

Replay DA in the same sequential proposer order. If all full-prefix exposure statuses agree, the next buyer finds the same available choice. The seller makes the same comparison. Induction preserves every tentative match.

For a failed component C with edge set E_C, let $P = (\Pi(M) \cap E_C) \setminus E_0$. If the current exposure is v^t, add

$$\sum_{e \in P : v_e^t = 0} v_e + \sum_{e \in P : v_e^t = 1} (1 - v_e) \geq 1.$$

Every successful plan must change one of these statuses, or the same local outcome and its blocking pair persist.

An unmatched buyer contributes its whole acceptable list. Full-market components are independent for matching, even when firms couple their audit decisions.

4. Exactness and stopping

Let $\mathcal{F}_C$ be the firms with clients in component C. Every successful plan stays feasible for the master. Each failure excludes a new audit assignment.

Whole-assignment method:	$2^{\lvert\mathcal{F}\rvert} + 1$ master solves at most,
Local methods:	$1 + \sum_{C} 2^{\lvert\mathcal{F}_C\rvert}$ at most.

A verified master candidate is optimal because its cost equals a valid lower bound. The same is true when that bound reaches a verified incumbent.

On interruption, use the master's certified dual bound as L. A candidate supplies U only after DA passes the full-market check.

The theorem assumes exact master optimization and exact oracle evaluation. Implemented MILP bounds have numerical tolerances. The bounds can be exponential.

5. Two solves versus exponentially many failures

The shared-firm instance has a unique support optimum $\{F_A, F_B\}$. Its first failed response produces N constraints

$$(1 - v_{a_i x_i}) + (1 - v_{b_i y_i}) + v_{c_i x_i} \geq 1.$$

Any competitive plan contains F_A, F_B, so these reduce to $v_{c_i x_i} = 1$. The second master therefore pays one private repair per copy and certifies $N + 2$.

Every set $\{F_A, F_B\} \cup \{f(c_i) : i \in T\}$ with $T \subsetneq [N]$ supports stability, but DA fails and its cost is below $N + 2$. There are $2^N - 1$ such sets. Whole-assignment cuts must remove each separately.

Both methods start with the support master and a verified upper bound. The comparison excludes initialization rounds and applies only to the specified cut families.

6. A sufficient certificate can cost too much

Let $M^* = \mathrm{DA}_B(E)$. Exposing every edge in $\Pi(M^*) \setminus E_0$ reproduces M^* by replay. Paying for these edges is vertex cover on endpoint firms.

But in this two-by-two component,

$$a : s \succ r, \quad b : s \succ r, \quad s : a \succ b, \quad r : b \succ a,$$

$E_0 = \{as, br\}$ already gives M^*, while the full prefix contains hidden bs. That extra disclosure is unnecessary for the outcome.

Take N private copies and one separate switch needing one audit. The true optimum is one. Prefix cover costs $N + 1$.

This also separates prefix completeness from reproducing M^* itself. It limits the certificate's cost, not its correctness.

7. Two other tractable cases

Switch markets with small portfolios. With private sellers, unit costs, distinct endpoint firms, and no repeated firm pair, two clients per buying firm give a degree-two firm graph. Vertex cover on its paths and cycles is linear-time.

Bounded interaction width. For each full-market component C, tabulate the local success predicate $\phi_C(x|_{\mathcal{F}_C})$. Make $\mathcal{F}_C$ a clique in a graph of firms. Given a width-t tree decomposition, Boolean constraint DP takes

$$O\left(\sum_C 2^{|\mathcal{F}_C|}\,\mathrm{poly}(|C|) + 2^{t+1}\,\mathrm{poly}(|I|)\right).$$

The first result applies only to the switch subclass. The second uses standard constraint dynamic programming. Here $|I|$ is input length. A component containing many firms can make t large.

8. Why simple repair rules can fail

Auditing a blocker's endpoints is not necessary. In the switch, qy initially blocks. Auditing p's firm removes the block by producing py, qx, although that firm represents neither q nor y.

Blocker reduction need not be submodular. Let a, b both rank $x \succ y$, and x, y both rank $a \succ b$. Start with $E_0 = \{bx\}$ and private firms.

Audited buyers	$\varnothing$	a	b	a, b
Number of blockers	2	1	2	0

For reduction $g(A)$, $g(a) + g(b) = 1 < 2 = g(a, b) + g(\varnothing)$.

9. A separate model for partnership bonuses

All pairs in E are available, but recommendation scores change:

$$\widetilde{u}_i(e) = u_i(e) + \lambda_i F_e, \qquad F_e \in \{0,1\}, \quad \lambda_i \geq 0.$$

Both endpoints share the same partnership label F_e. Scores and bonuses are rational. Let P, Q be the strict base and modified rankings, with the outside option last.

Define $x \succ_i^* y$ when i prefers x to y in *both* rankings. Then

$$\text{stable under both } P, Q \quad \Longleftrightarrow \quad \text{super-stable under } \succ^* .$$

Existence and construction are polynomial-time by partial-order super-stability.

This is a separate model from the access problem. The shared binary label enables the reduction to Cseh and Juhos (2021).

10. Why the binary label matters

For a candidate pair e and endpoint i, let e_i^M be its current match. Give the unmatched option label zero. The gain from deviating satisfies

$$\Delta_i^Q(e) = \Delta_i^P(e) + \lambda_i(F_e - F_{e_i^M}).$$

If $F_e = 1$, both endpoints' gains weakly increase. Stability in Q supplies an endpoint that rejects e in Q, and therefore in P too.

If $F_e = 0$, both gains weakly decrease. Use the rejecting endpoint from P. Thus every unmatched pair has one endpoint rejecting it under both profiles, which is exactly super-stability of the intersection orders.

Conversely, a common rejecting endpoint prevents blocking in each profile. The same rejection holds for independent strengths $\theta_i \in [0, \lambda_i]$ because each score difference is affine.

11. The bonus result has limits

A simultaneous solution need not exist. In a two-by-two market, give everyone a base score of one for their diagonal partner and zero for the other. Give crossed pairs a bonus of two. The two profiles have different unique stable matchings.

Three label levels also break the binary reduction. Take $M = \{ar, ts\}$ and candidate as, with

$$F_{ar} = 2, \quad F_{as} = 1, \quad F_{ts} = 0, \qquad \lambda_i = 2.$$

Set $u_a(r) = 0, u_a(s) = 1$ and $u_s(a) = 0, u_s(t) = 1$. Seller s rejects as under base scores. Buyer a rejects it after bonuses. There is no common rejecting endpoint.

This disproves that reduction for three levels. It does not prove NP-hardness. Use the three-edge acceptable graph $\{ar, as, ts\}$, with sole partners for r, t.

12. What was recorded and checked

- 160 core, 32 larger, and 32 cost/portfolio markets. The 40 ablation runs reuse core instances.
- Eight constructed response-gap sizes and five prefix-cover-gap sizes, kept separate from random-market results.
- 1,280 core method outputs and 614 optimal plans independently replayed. DA, prefix replay, reductions, and small optima checked against enumeration.
- CPU runs on ARM64 macOS, one HiGHS thread, fixed seeds, rotated method order, source hashes, and explicit time caps.

95% intervals use 20,000 bootstrap replicates over 20 shared seed blocks. All final core matchings are perfect because the generator uses balanced complete blocks. That is a generator property, not evidence about housing scarcity.

13. A common priority removes the gap

Suppose all sellers use the same strict priority over buyers. There is a unique stable matching M^*: clients choose in priority order.

$$M^* \subseteq H \subseteq E \quad \Longrightarrow \quad \mathrm{DA}_B(H) = M^*.$$

In any visible graph H containing M^*, each full-market choice remains available at its turn. Hence

$$C_{\text{support}} = C_{\text{DA}}.$$

We just need to cover the hidden edges of M^* with endpoint-firm audits. For side-exclusive firms, weighted bipartite vertex cover solves this by minimum cut.

A consequence of serial dictatorship. The minimum-cut step requires side-exclusive firms. Other tractable cases appear in Appendix 7.

14. Proposed listing study

What we could observe

Rents and features in a fixed listing catalogue.

Listing IDs recommended by a fixed AI model.

Census household distributions for generating fictional profiles.

What would remain modeled

Preferences on both sides.

Agent portfolios and audit costs, unless independently verified.

The matching rule and its resulting allocations.

Hold catalogue and recommendation count fixed. Randomize listing order, repeat prompts, and compare audit policies on the same markets.

Proposed, not run. Candidate sources: Two Sigma rental listings and American Community Survey microdata. Verify dates and manager-ID meanings. Inquiry interest is not a preference list. Do not link Census records to individual listings.

15. The cost of an access target

Fix each modeled buyer's affordable feasible options F_b. Let G^+ be lower-income households with at least one such option.

$$\mathrm{Access}(A) = \frac{1}{|G^+|} \sum_{b \in G^+} \frac{|\{s \in F_b : bs \in E(A)\}|}{|F_b|}.$$

Let $C(\tau)$ be the cheapest plan that makes DA full-market stable and achieves $\mathrm{Access}(A) \geq \tau$. Study

$$\Delta C(\tau) = C(\tau) - C(0).$$

How does that extra cost change with scarcity, firm concentration, and the recommender that chooses E_0?

A proposed extension, not a completed result. Assume $|G^+| > 0$ and $\tau \in [0, 1]$. Report households with no affordable option separately. Access measures options, not eventual leases.

16. Matching and optimization references

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